CSC311 project

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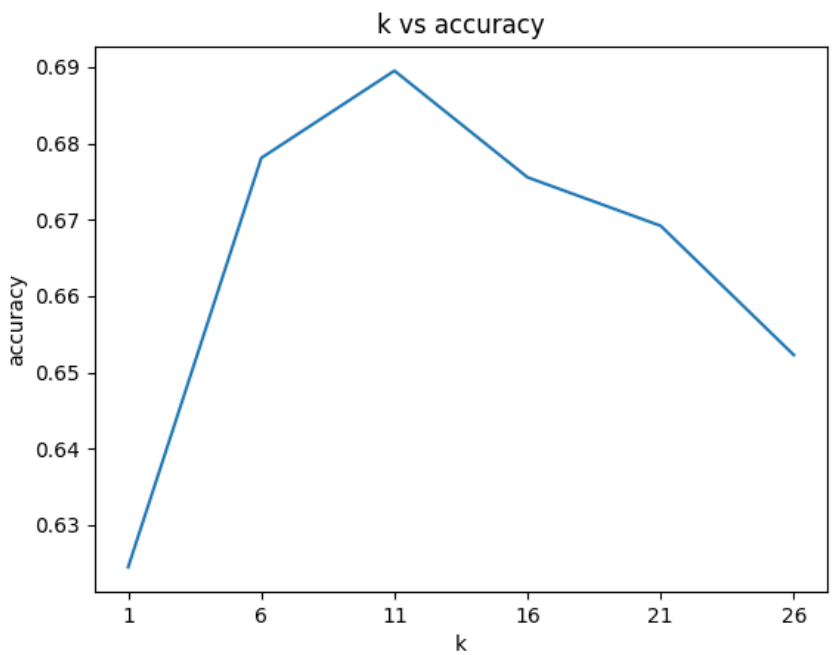
Part I

1 k-Nearest Neighbor

(a) Validation data accuracy as a function of k



(b) We choose $k^* = 11$. Final test accuracy:

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Accuracy: 0.6841659610499576
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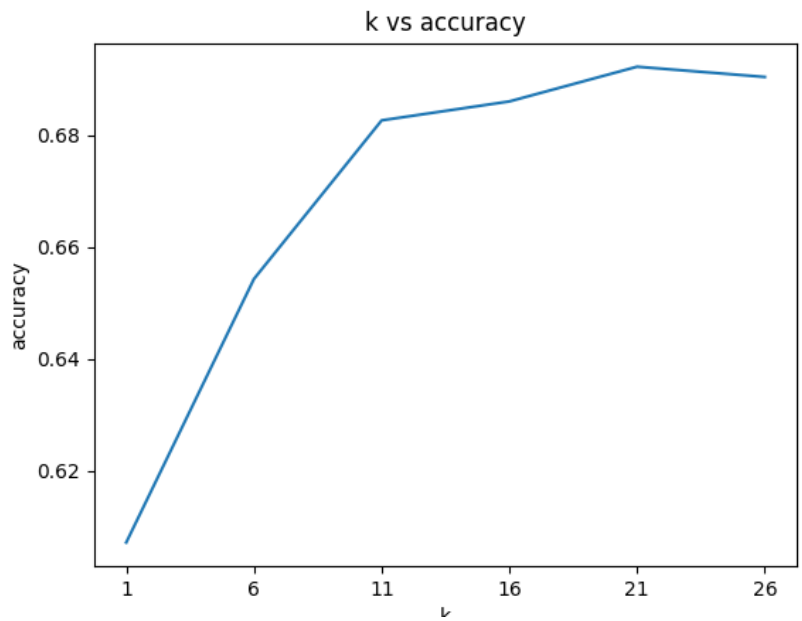
(c) In item-based collaborative filtering, the assumption is that if a group of students answers one question similarly, then a new question answered by the same group is likely to have a similar level of correctness as the previous question.



We choose $k^* = 21$. Final test accuracy:

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Accuracy k = 21 : 0.6816257408975445
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(d) Test accuracy for user-based filtering is higher

(e) 1. Curse of dimensionality 1774 questions means a dimension of 1774, which is very high. It is possible kNN is no longer effective at numbers this high.

2. Assumptions The assumptions given about how students with similar answers will answer similarly for other questions, or that similar groups of students will have a similar level of correctness may not necessarily hold. Some students may be better at certain questions than others, while all of them may do well at easy questions. In this case, the assumption would not hold and the algorithm would not be effective.

2 Item Response Theory

(a) Let N be the total number of students, and M be the total number of questions.

The likelihood:

$$
\begin{aligned}
p(\boldsymbol{C}|\boldsymbol{\theta},\boldsymbol{\beta}) &= \prod_{i=1}^{N}\prod_{j=1}^{M} p(c_{ij}|\theta_i,\beta_j) \\
&= \prod_{i=1}^{N}\prod_{j=1}^{M} p(c_{ij}=1|\theta_i,\beta_j)^{c_ij} p(c_{ij}=0|\theta_i,\beta_j)^{1-c_{ij}}
\end{aligned}
$$

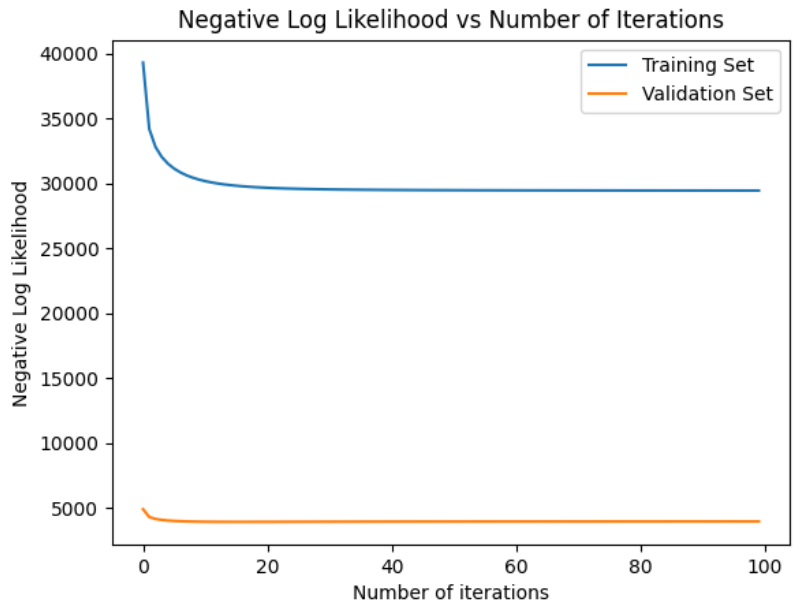
Log-likelihood:

$$
\begin{aligned}
L(\boldsymbol{C}|\boldsymbol{\theta},\boldsymbol{\beta}) &= \log p(\boldsymbol{C}|\boldsymbol{\theta},\boldsymbol{\beta}) \\
&= \sum_{i=1}^{N}\sum_{j=1}^{M} c_{ij}\log p(c_{ij}=1|\theta_i,\beta_i) + (1-c_{ij})\log p(c_{ij}=0|\theta_i,\beta_i) \\
&= \sum_{i=1}^{N}\sum_{j=1}^{M} c_{ij}\log\frac{exp(\theta_i-\beta_j)}{1+exp(\theta_i-\beta_j)} + (1-c_{ij})\log\frac{1}{1+exp(\theta_i-\beta_j)} \\
&= \sum_{i=1}^{N}\sum_{j=1}^{M} c_{ij}(\theta_i-\beta_j) - c_{ij}\log\left(1+exp(\theta_i-\beta_j)\right) + (c_{ij}-1)\log\left(1+exp(\theta_i-\beta_j)\right) \\
&= \sum_{i=1}^{N}\sum_{j=1}^{M} c_{ij}(\theta_i-\beta_j) - \log\left(1+exp(\theta_i-\beta_j)\right)
\end{aligned}
$$

Derivatives:

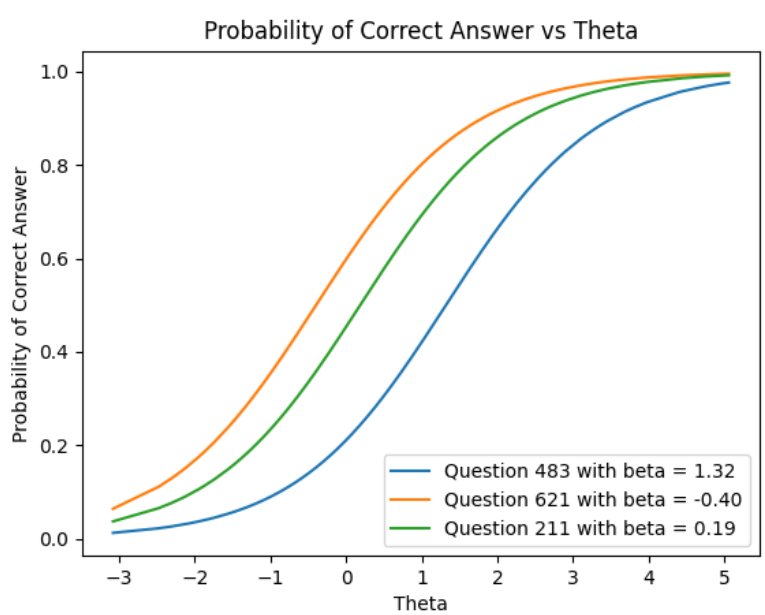
$$
\begin{aligned}
\frac{\partial L}{\partial \theta_i} &= \sum_{j=1}^{M} c_{ij} - \frac{exp(\theta_i-\beta_j)}{1+exp(\theta_i-\beta_j)} \\
\frac{\partial L}{\partial \beta_j} &= \sum_{i=1}^{N} -c_{ij} + \frac{exp(\theta_i-\beta_j)}{1+exp(\theta_i-\beta_j)}
\end{aligned}
$$

(b) Chosen hyperparameter: learning rate = 0.02, number of iteration = 100



(c) Validation Accuracy: 0.7064634490544736; Test Accuracy: 0.7090036692068868

(d) The shape of the curves all look like sigmoid functions, since the probability is defined as $p(c_{ij} = 1|\theta_i, \beta_j) = \frac{exp(\theta_i - \beta_j)}{1+exp(\theta_i - \beta_j)}$. In this model, theta θ reflects a student's ability, while β reflects question difficulty.The probability of a correct answer increases with a student's ability level—this is shown by the upward trend of the curves. Furthermore, for a given θ, the curve with the lower β is higher, indicating that questions of lower difficulty have a higher likelihood of being answered correctly by students across all ability levels. The graphical interpretation aligns with the theoretical expectations that easier questions will have higher probabilities of correct responses.



3 Neural Networks

(a) Alternating Least Squares in an optimization algorithm used to factorize a matrix into two matrices which can then be used to make predictions. Neural Networks are machine learning models with nodes organized in layers, used to make predictions on classification and regression problems. These models do use optimization algorithms to optimize their architecture, weights, and other parameters. Thus one major difference is the purpose of these two things, as ALS is only an optimization algorithm while Neural Networks has a much broader function. We can see this as the Autoencoder model is useful for a much broader range of problems than ALS.

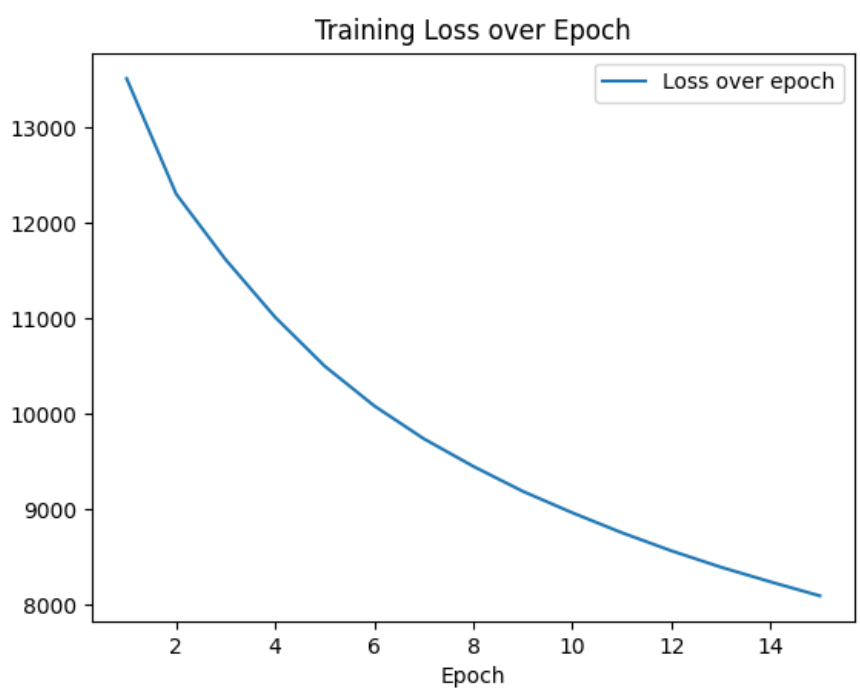
A second difference is the Neural Networks simultaneously optimizes all the parameters with backpropagation and gradient descent. This is different from ALS as ALS switches between a fixed parameter and optimizing the other until all parameters are optimized.

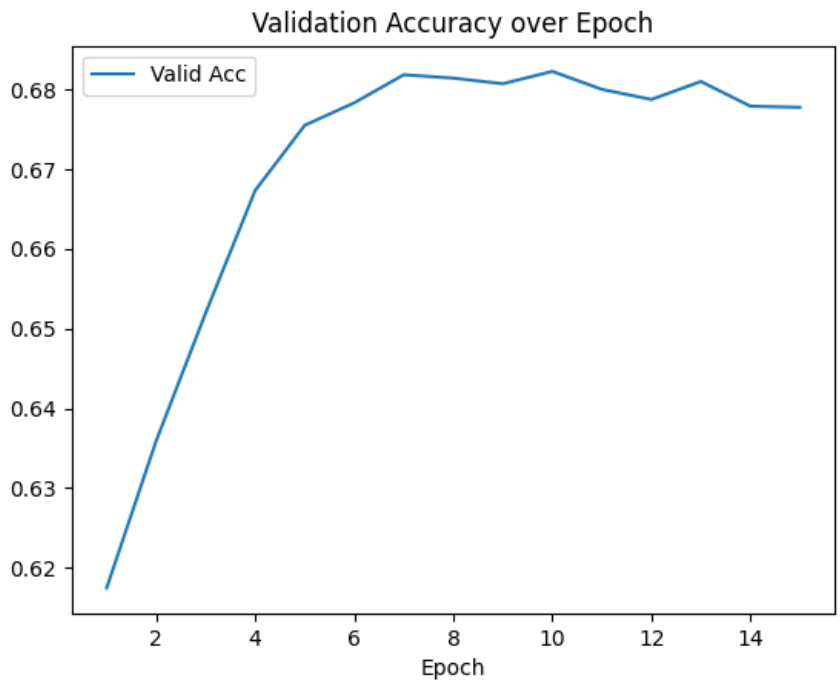
Another difference is the ALS algorithm reduces the dimension of the data and keeps the data represented in this low dimension for predictions. Autoencoders encode the data to a lower dimension and then decode the data back to the original dimension. Then $\tilde{\mathbf{x}}$ is used to make predictions.

A final difference is that neural networks and more specifically autoencoders can actually project the data onto a nonlinear manifold, which allows for powerful codes for any given dimension. ALS is not primarily used for problems that need non-linear dimensionality reduction.

(c) We set $k^* = 10$, learning rate $= 0.1$, and epoch count to 15 after tuning.

(d) Graphs Below:





Final Accuracy 68.50%.

(e) We set $\lambda = 0.001$. The accuracy was 68.61%. The regularization penalty increased the training loss which was expected. I do not believe that it made a great impact, and the model performs about the same with or without it. This may be because we are working with a relatively simple model (complexity-wise), where regularization does not usually have a great impact.

4 Ensemble

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Average Validation accuracy:0.6984664596857654
Average Test accuracy:0.6996895286480384
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We used IRT as the base model for all 3. We performed bootstrapping by randomly sampling the training set with replacement to get bootstrap sets of the same size as the training set. Then, we train and predict the model the same way as in Q2, and take the average accuracy over the 3 models.

Compared to our accuracy in Q2, the validation accuracy was 0.8 percent lower and the test accuracy was 0.9 percent lower. This is very close to the accuracy in Q2, but it being slightly lower could possibly be explained by the nature of sparse matrices, if one point is not sampled at all it could mean an entire question would have nothing to train on at all.

Part II

1. Formal Description

We propose an extension to the traditional autoencoder model by employing a Variational AutoEncoder (VAE). Unlike standard autoencoders that aim to minimize reconstruction errors, VAEs introduce a probabilistic graphical model with latent variables. The key equations governing our VAE are:

$$\text{Encoder: } q_\phi(z|x) = \mathcal{N}(z; \mu_\phi(x), \sigma_\phi^2(x)) \tag{1}$$

$$\text{Decoder: } p_\theta(x|z) = \mathcal{N}(x; \mu_\theta(z), \sigma_\theta^2(z)) \tag{2}$$

$$\text{Loss Function: } \mathcal{L}(\phi, \theta; x, z) = \text{BCE}(x, \mu_\theta(z)) + \text{KL}(q_\phi(z|x)||p(z)) \tag{3}$$

Here, ϕ and θ are the parameters of the encoder and decoder networks, respectively. BCE represents the binary cross-entropy, and KL denotes the Kullback-Leibler divergence.

The expected improvement from this model is twofold:

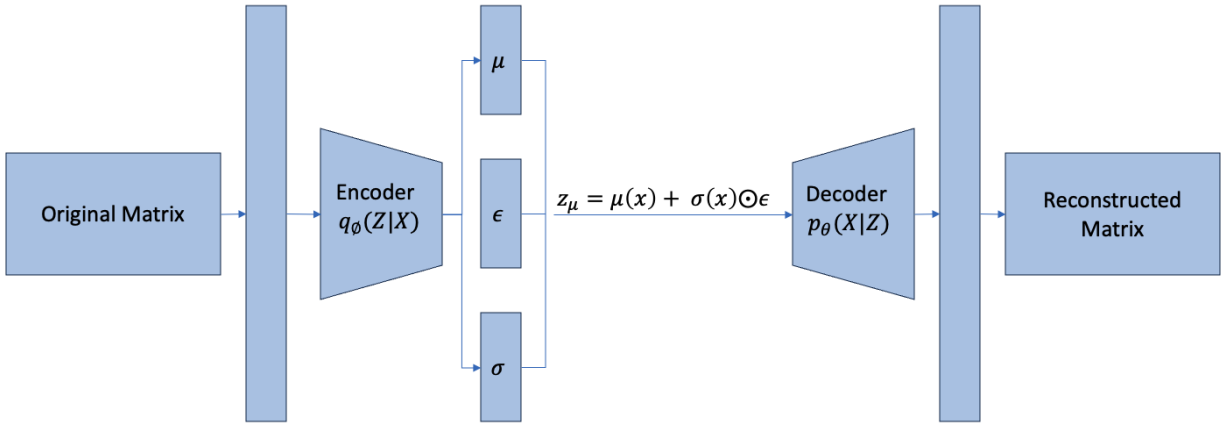
1. **Optimization:** By capturing the data distribution in the latent space, the VAE is likely to learn more robust features, enhancing optimization.
2. **Regularization:** The KL divergence term in the loss function acts as a regularizer, potentially reducing overfitting.

2. Figure or Diagram

Below is a figure detailing the modifications in comparison with standard AutoEncoder

Feature	**Standard AutoEncoder**	**Variational AutoEncoder**
Latent Space Representation	Deterministic	Probabilistic
Encoder Output	Encoded Representation	Latent Distribution (μ, σ)
Decoder Input	Encoded Representation	Sampled Latent Variable (z)
Reconstruction Loss	MSE or similar	Binary Cross-Entropy (BCE)
Regularization	None	Kullback-Leibler (KL) Divergence
Optimization Objective	Minimize Reconstruction Error	Minimize BCE + KL Divergence

Table 1: Comparison between Standard AutoEncoder and Variational AutoEncoder

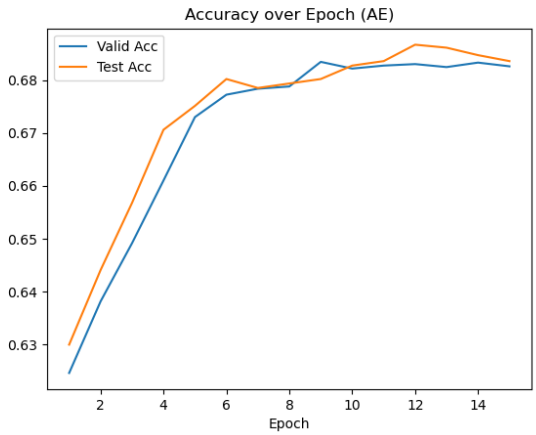


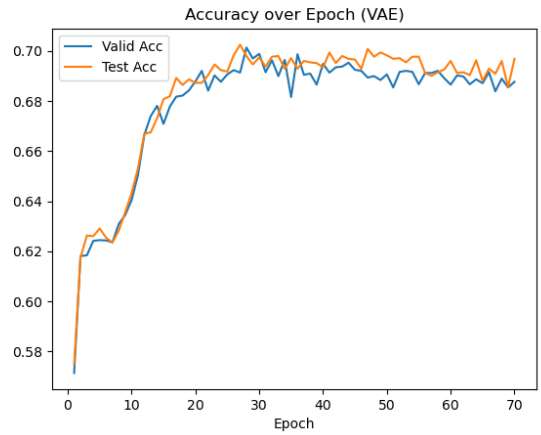
Variational Autoencoder (VAE) applies a probabilistic approach to dimensionality reduction and generative modeling. In this VAE, an encoder network first compresses the input data (Original students & questions Matrix) into a lower-dimensional latent space, characterized by statistical parameters—mean and variance. This step transforms the data into a distribution from which latent variables are sampled, using a method known as the "**reparameterization trick**." Here, an element-wise product combines the variance with a **random noise vector** to introduce stochasticity, allowing gradient-based optimization through backpropagation. The decoder network then reconstructs the input data from these latent variables, aiming to produce a Reconstructed Matrix that closely approximates the Original Matrix. The entire process enables the VAE to learn efficient data representations and generate new samples that mimic the original dataset.

3. Comparison or Demonstration

(1) Comparison with Base Model

Model	Latent Dimension	Learning Rate	Epoch	Validation Accuracy	Test Accuracy
Auto Encoder	10	0.1	15	68.85%	67.99%
VAE	5	0.01	50	69.01%	69.01%





We compared the Variational Autoencoder (VAE) model with the original Autoencoder (AE) model. The AE, with a higher latent dimension of 10 and a larger learning rate of 0.1, was trained for 15 epochs, achieving validation and test accuracies of 68.85% and 67.99%, respectively. In contrast, the VAE, with a latent dimension of 5 and a learning rate of 0.01, underwent training for 50 epochs, resulting in both validation and test accuracies of 69.01%. The accuracy trend graphs reveal that the AE rapidly reached peak accuracy and then stabilized, while the VAE demonstrated a gradual improvement with more pronounced fluctuations throughout its training epochs, suggesting a nuanced learning process with slightly superior performance despite a more compact latent representation and extended training duration.

(2) Experiment

Our study posited that incorporating Kullback-Leibler (KL) divergence as a regularization term in the loss function would mitigate overfitting, thereby enhancing the model's accuracy. To empirically validate this hypothesis, we structured an experimental setup that allowed us to measure the effects of KL divergence on model performance.

Model	Latent Dimension	Learning Rate	Epoch	Validation Accuracy	Test Accuracy
With KL divergence	5	0.01	50	69.01%	69.01%
Without KL divergence	5	0.01	50	67.74%	67.15%

The experiment's results, as depicted in the table, illustrate a clear comparison between two models: one incorporating Kullback-Leibler (KL) divergence in its loss function and one without. Both models were configured with an identical latent dimension of 5 and a learning rate of 0.01, and both were trained over the same number of epochs (50).

The model with KL divergence regularization achieved validation and test accuracies of 69.01%, while the model without KL divergence showed lower accuracies of 67.74% for validation and 67.15% for test. These results empirically support the initial hypothesis that the inclusion of KL divergence as a regularization term can effectively reduce overfitting, enhancing model accuracy.

4. Limitations

While the Variational AutoEncoder (VAE) model presents improvements in certain aspects, there are some limitations:

(1) Performance in High-Dimensional Sparse Data

Variational Autoencoders (VAEs) may underperform when processing datasets that are extensive in dimension yet contain many unobserved elements. The challenge arises from the VAE's insufficient capacity to discern the probabilistic distributions when the latent space is dominated by sparse information. The lack of dense data can hinder the VAE's ability to learn meaningful distributions, particularly when the input data is predominantly composed of zero values, which diminishes the strength of the learning signal.

In the context of predicting student's answer correctness, where the data matrix is constructed from students' responses to a set of questions, the incomplete nature of this matrix can impair the VAE's predictive accuracy. Given that a student might have only attempted a fraction of the available questions, the model's predictive power for that student's performance on unattempted questions is compromised.

A possible future improvement could include the adoption of priors that are conducive to sparsity within the latent space or the exploration of alternative models that are designed to handle sparse datasets more efficiently.

(2) Balancing Reconstruction and Regularization

In our objective function, there are two terms: binary cross-entropy loss and KL divergence. The binary cross entropy drives the accuracy of the reconstruction, while the KL divergence promotes regularization. Striking the right balance between these elements is crucial, as over-prioritizing either can degrade the model's effectiveness.

For instance, disproportionately weighting the KL divergence can lead to a model that is too generalized, failing to capture the specific patterns in student responses. Conversely, if the reconstruction loss dominates, the model may overfit, excelling on training data but faltering on unseen questions.

Potential solutions may include adaptive methods that dynamically adjust the importance of each term throughout training or incorporating cutting-edge techniques in variational inference that refine the balance between reconstruction loss and the generality of the learned model.

Appendix

Contribution

Part A

Q1: Michael Gao

Q2: Zeyu Zhang

Q3: Humza Iqbal

Q4: Michael Gao

Part B

Discussion: Michael Gao, Zeyu Zhang, Humza Iqbal

Code Implementation: Zeyu Zhang

Report Write-up: Michael Gao, Zeyu Zhang, Humza Iqbal

Other Notes

While looking for ways to improve the algorithms from Part A for Part B, we were able to find Elastic Net Regularization. This regularization term is able to be implemented in both the IRT and the Neural Network. Elastic Nets uses 2 parameters, α and λ, to create a tradeoff between LASSO and RIDGE regression, where $\alpha, \lambda \in [0, 1]$.

$$\min_{\theta} \sum_{\mathbf{v} \in S} \|\mathbf{v} - f(\mathbf{v}; \theta)\|_2^2 + \lambda\big[(1-\alpha)/2(\|\mathbf{W}^{(1)}\|_F^2 + \|\mathbf{W}^{(2)}\|_F^2) + \alpha(\|\mathbf{W}^{(1)}\|_F + \|\mathbf{W}^{(2)}\|_F)\big]$$

This would be the modified objective function from Q3. The parameter α controls how much of each LASSO or RIDGE we are using in the model. The parameter λ controls how strong the overall regularization term is. The L1 regularization term is well-suited for sparse data, as it allows some of the covariates to be sent to 0. The L2 regularization term strengthens the model's robustness against Multicollinearity, and we assume the data in this case may exhibit multicollinearity. For example, if there are redundant questions. After some tuning, we were not able to improve our model with this setup, and our highest accuracy was 68.2%.

References

Hastie, T., Qian, J., & Tay, K. (2023, March 27). *An introduction to 'glmnet'*. glmnet https://glmnet.stanford.edu/articles/glmnet.html